

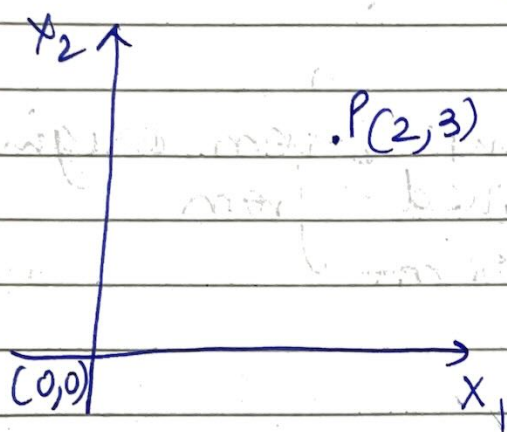
Module 2: ch-2 Linear Algebra

V2: Introduction to vectors (2D, 3D, nD)

Row Vector and Column Vector:-

Point/vector

In a 2^{D} coordinate system, let's have axis x_1 and x_2 for x-axis and y-axis.



Point p has x-coordinate as 2 and y-coordinate as 3

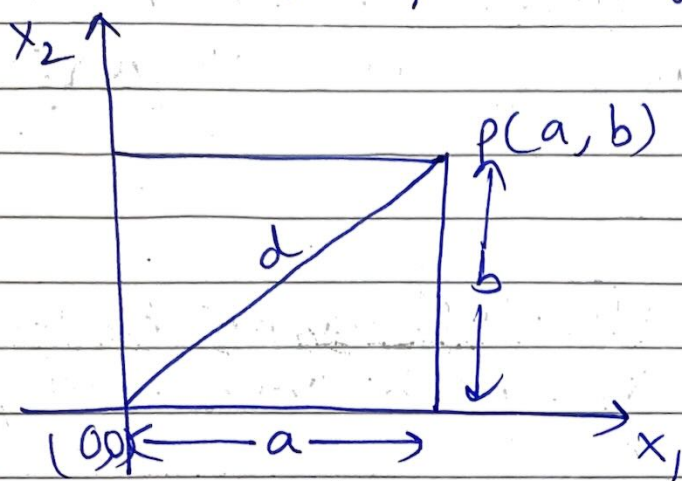
$p[2,3]$. Here 2 is x_1 component of vector and 3 as x_2 component of vector.

In 3-D q point has $q = [2, 3, 5]$ where 2 is x_1 component, 3 is x_2 component and 5 as x_3 component.

For n -D, we have n -dim point, where its vector is represented as $x = [2, 3, 4, \dots, n]$ where 2, 3, 4, ..., n are its components.

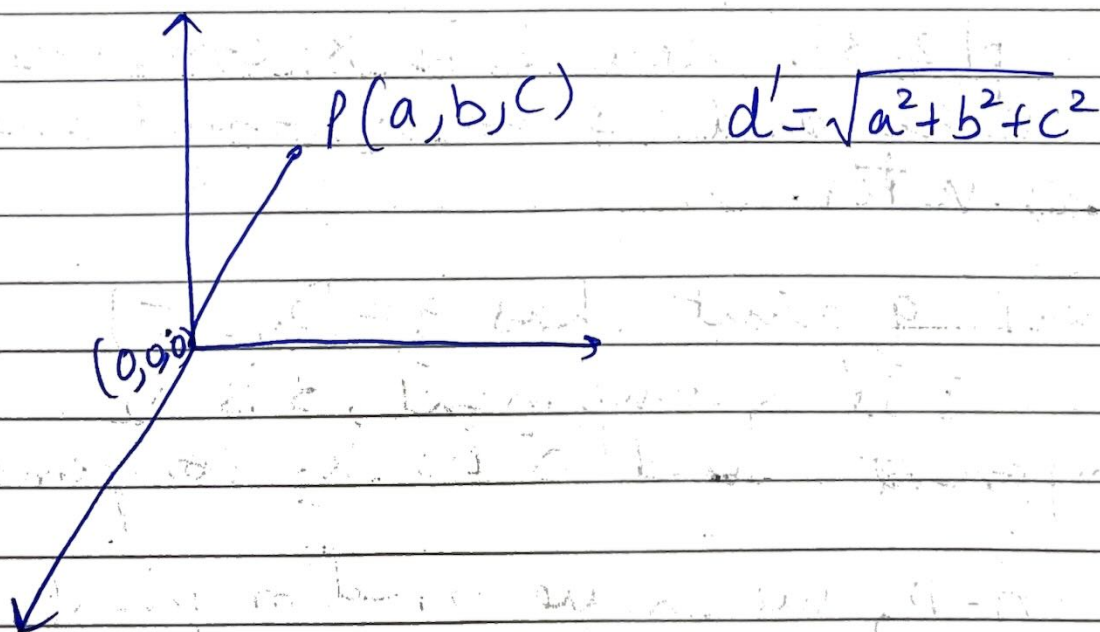
distance

Distance of a point from origin



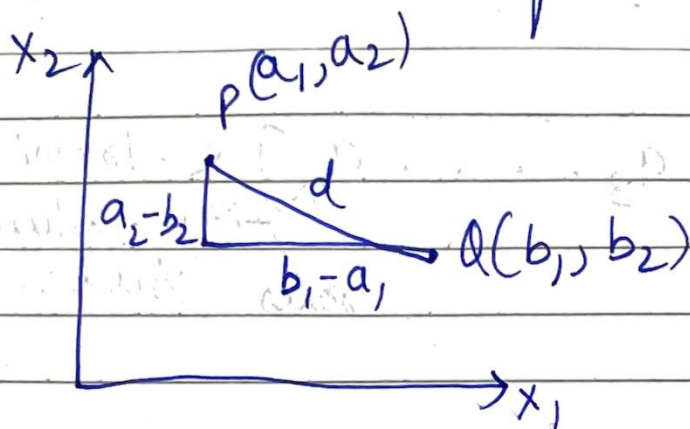
d = distance of point from origin.
It can be derived from
Pythagoras theorem.

$$d = \sqrt{a^2 + b^2}$$



For $nD \Rightarrow d = \sqrt{a_1^2 + a_2^2 + \dots + a_n^2}$
if $P = [a_1, a_2, \dots, a_n]$

Distance between 2 points in 2D:-



$$d = \sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2}$$

In 3D:-

$$P(a_1, b_1, c_1) \quad Q(a_2, b_2, c_2)$$

$$d = \sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2 + (a_3 - b_3)^2}$$

In n-D space,

$$P(a_1, a_2, \dots, a_n) \quad Q(b_1, b_2, \dots, b_n)$$

$$d_{pq} = \sqrt{\sum_{i=1}^n (a_i - b_i)^2}$$

Row vector:

$A = [a_1, a_2, a_3, \dots, a_n]$ ← row n columns
 1xn-column
 row subscript

Column Vector

$$B = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

$n \times 1$
 rows column
 subscript

Matrix is represented as $A_{m \times n}$

$$\begin{matrix} 1 \\ 2 \\ \vdots \\ m \end{matrix} \left[\begin{array}{c} \\ \\ \\ \\ \end{array} \right] \quad \begin{matrix} N \\ \\ \\ \\ \end{matrix}$$

$m \times n$

Matrix = Array
of arrays.

V3: Dot product and angle between 2 vectors.

$$A = [a_1, a_2 \dots a_n]$$

$$B = [b_1, b_2 \dots b_n]$$

Addition of 2 vectors +

$$C = A + B = [a_1 + b_1, a_2 + b_2, \dots a_n + b_n]$$

Multiplication of 2 vectors :-

1) Dot product ($a \cdot b$)

2) Cross product.

$$a \cdot b = a_1 b_1 + a_2 b_2 + \dots + a_n b_n$$

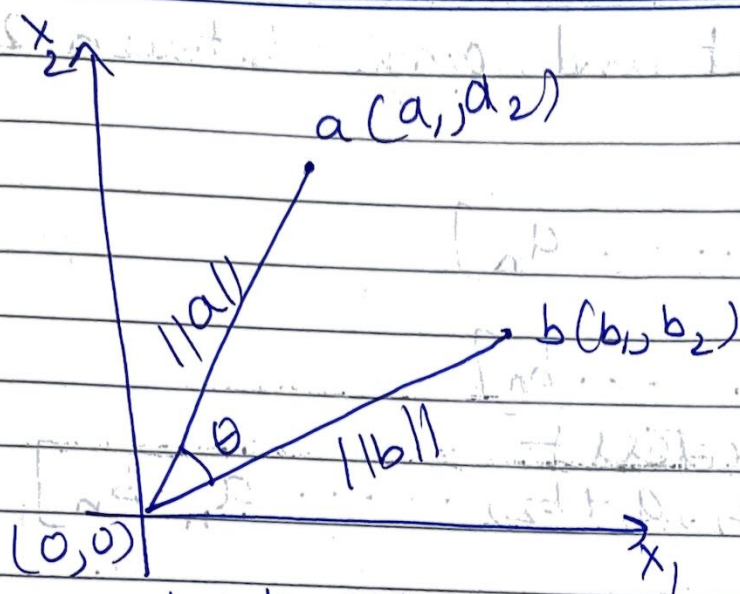
$$= [a_1, a_2 \dots a_n] \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

$$= a^T \cdot b = \sum_{i=1}^n a_i b_i$$

Vector is by default a column vector.

$$a = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$$

$$a^T = [a_1, a_2 \dots a_n]$$



$$a \cdot b = \|a\| \|b\| \cos \theta$$

Length
of
 a

Length of a = Distance of a from origin -
 " " b = " " " " b " "

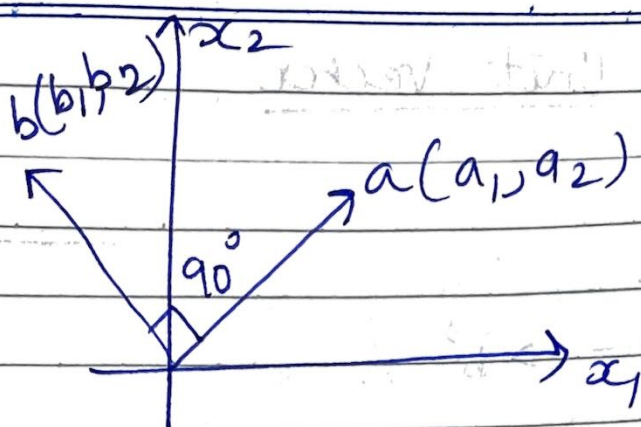
$$a \cdot b = a_1 b_1 + a_2 b_2$$

$$= \|a\| \|b\| \cos \theta$$

$$\theta = \cos^{-1} \left\{ \frac{a_1 b_1 + a_2 b_2}{\|a\| \|b\|} \right\}$$

We know.

$$\|a\| = \sqrt{a_1^2 + a_2^2}$$



$$a \cdot b = \|a\| \|b\| \cos 90^\circ$$

$$a \cdot b = 0$$

If dot product b/w 2 vectors is 0, then they are perpendicular to each other.

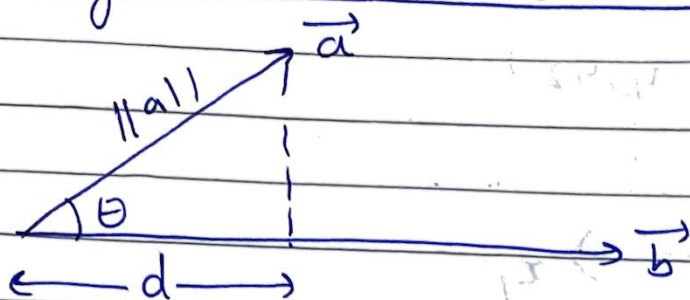
$$\text{So, } \theta_{a,b} = \cos^{-1} \left(\frac{\sum_{i=1}^n a_i b_i}{\|a\| \|b\|} \right)$$

$$a \cdot a = a_1 a_1 + a_2 a_2 + \dots + a_n a_n$$

$$= a_1^2 + a_2^2 + \dots + a_n^2$$

$$= \|a\|^2$$

V4: Projection and Unit vector



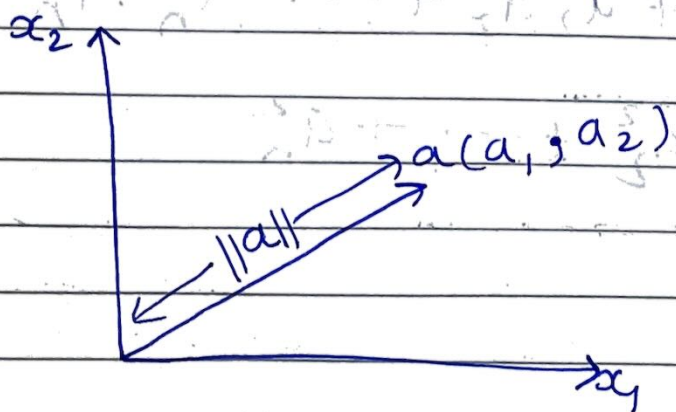
$d = \text{Projection of } \vec{a} \text{ on } \vec{b}$

But $d = ||a|| \cos \theta$ ——— (1)

And $\vec{a} \cdot \vec{b} = \sum_{i=1}^n a_i b_i = ||a|| ||b|| \cos \theta$

$$\text{So, } d = \frac{\vec{a} \cdot \vec{b}}{||b||} = \frac{||a|| ||b|| \cos \theta}{||b||}$$

Unit Vector :-

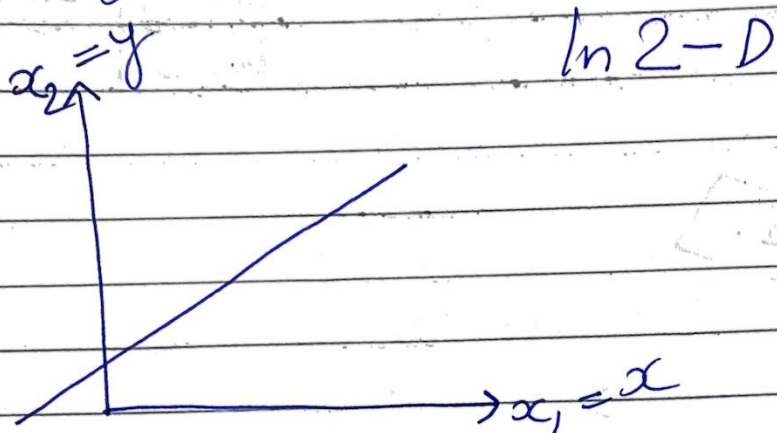


$$\hat{a} (\text{unit vector } a) = \frac{\vec{a}}{||a||}$$

$\hat{a}$ is in direction of $\vec{a}$.

$$\|\hat{a}\| = 1$$

V5: Equation of a line (2-D), Plane (3-D) and Hyperplane (n-D), Plane passing through origin, Normal to a plane.



$$y = mx + c$$

m = slope

c = y-intercept.

$y = mx + c$ can also be written as:-

$$ax + by + c = 0$$

$$y = \frac{-c}{b} - \frac{a}{b}x$$

$$ax_1 + bx_2 + c = 0$$

$$w_1x_1 + w_2x_2 + w_0 = 0 \text{ --- In 2-D.}$$

$$a = w_1$$

$$b = w_2$$

$$c = w_0 \text{ --- (1)}$$

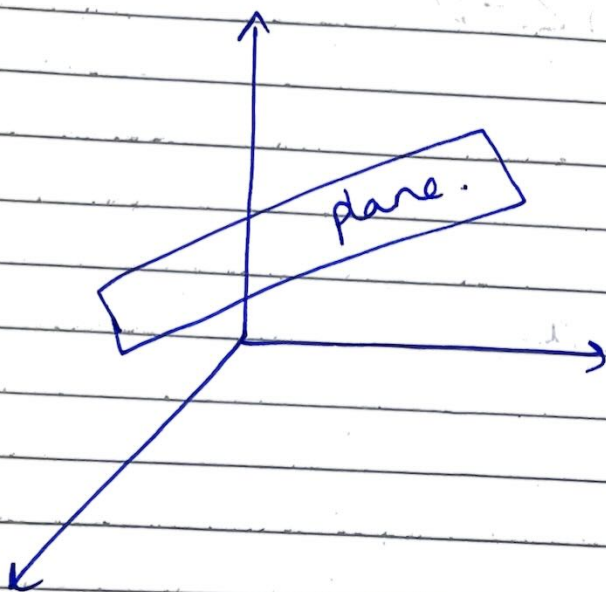
In 3-D

Eq. of plane:-

$$ax + by + cz + d = 0.$$

$$w_1 x_1 + w_2 x_2 + w_3 x_3 + w_0 = 0 \rightarrow$$

Eq. of plane
in 3-D.



In n-D

Eq. of Hyperplane:-

$$w_0 + w_1 x_1 + w_2 x_2 + \dots + w_n x_n = 0.$$

Sum notation

$$w_0 + \sum_{i=1}^n w_i x_i = 0.$$

Vector notation

$$w_0 + [w_1, w_2, \dots, w_n] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = 0$$

$$W_0 + W_{1 \times n} x_{n \times 1}$$

By default every vector is a column vector $\therefore$ So, $W_0 + W_{n \times 1} x_{n \times 1}$. Here W needs to be transposed

$$\boxed{W_0 + W^T x = 0}$$

(-sed)

From eq (1), we have

$$C = W_0$$

If line passes through origin $\therefore$

$$C = 0$$

$$\text{So, } W_0 = 0$$

And hence

$$\boxed{W^T x = 0}$$

Equation of plane passing through origin.

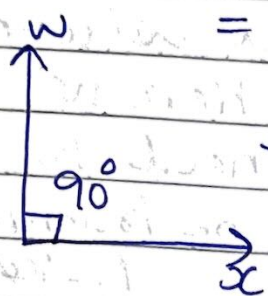
Let there be a plane Π in dimension

$$\Pi_n : W^T x = 0$$

$$W = \begin{bmatrix} W_1 \\ W_2 \\ \vdots \\ W_n \end{bmatrix} \quad x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

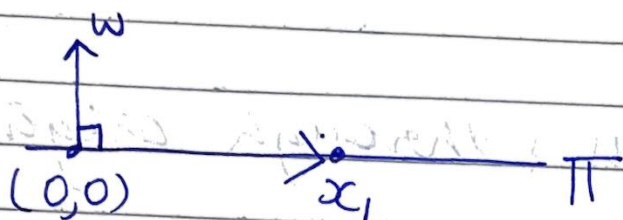
$$W \cdot x = W^T x$$

$$= \|W\| \|x\| \cos \theta_{w,x} = 0 \quad \text{--- Eq. of plane}$$



Then, $\boxed{\theta_{w,x} = 90^\circ}$

Hence $w \perp x$

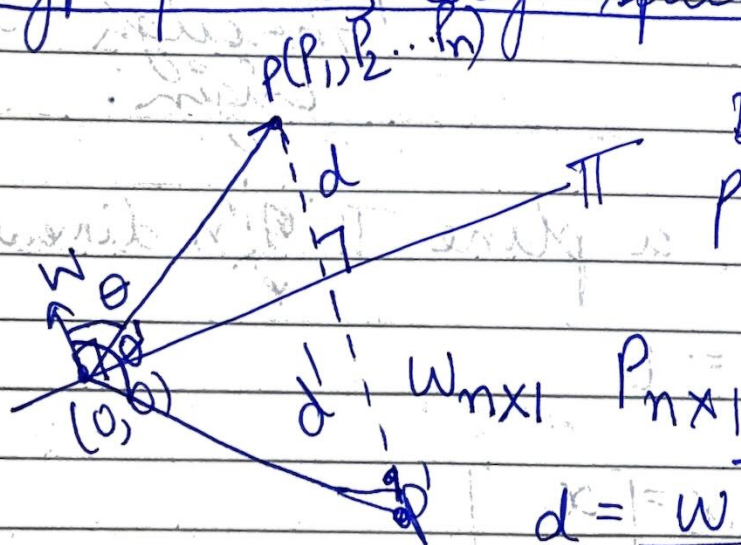


$$w \cdot x_i = 0$$

If w is $\perp$ to Π

Then $w^T x_i = 0 \quad \forall x_i \in \Pi$

V6: Distance of a point from a Plane/Hyperplane, Half spaces



Distance of a point in n -D.

$$d = \frac{w^T P}{\|w\|}$$

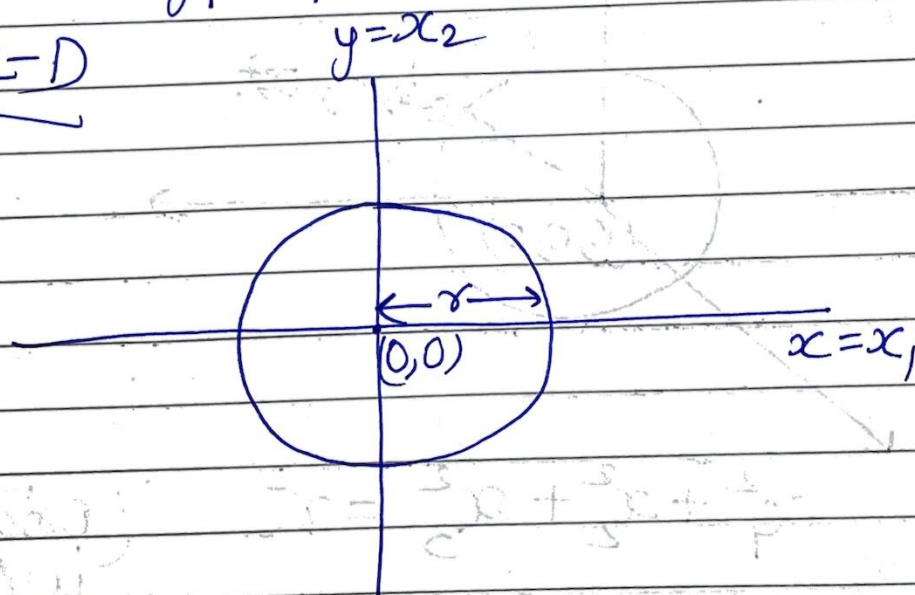
Since θ is acute and P & W lie on same ~~to~~ side, hence point P is true. $\|W\| \|P\| \cos(< 90^\circ) = +ve$

Since θ' is obtuse and P' and W lie in opp direction so P' is nega
 $\|W\| \|P'\| \cos(> 90^\circ) = -ve$

A hyperplane divides the region into 2-half spaces.

v7: Equation of a circle (2-D), Sphere (3-D) and Hypersphere (n-D)

In 2-D



$$x^2 + y^2 = r^2$$

$P(x_1, x_2)$

(centre is origin)

$C = (h, k)$

$$(x-h)^2 + (y-k)^2 = r^2$$

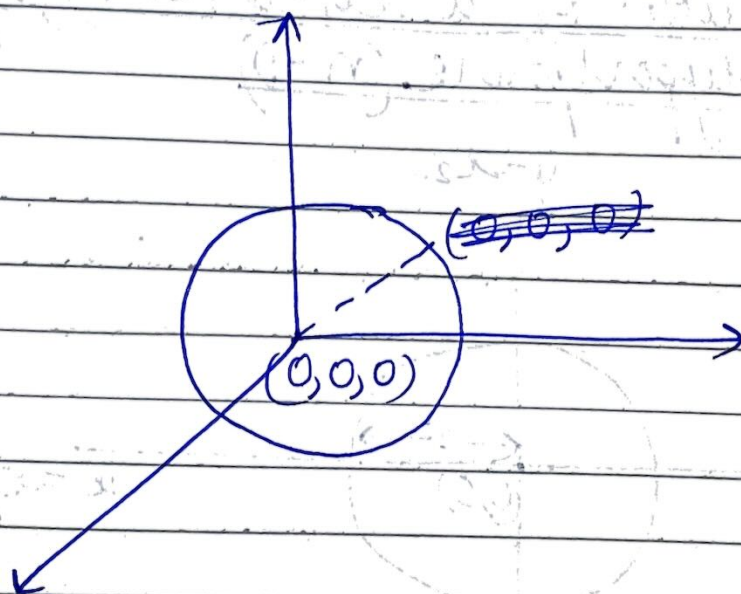
To determine if point lies inside or outside the $\odot$

If $x_1^2 + x_2^2 < r^2 \Rightarrow p$ lies inside

$x_1^2 + x_2^2 = r^2 \Rightarrow p$ lies on the $\odot$

$x_1^2 + x_2^2 > r^2 \Rightarrow p$ lies outside.

In 3-D



$$x_1^2 + x_2^2 + x_3^2 = r^2 \quad (\text{origin is centre})$$

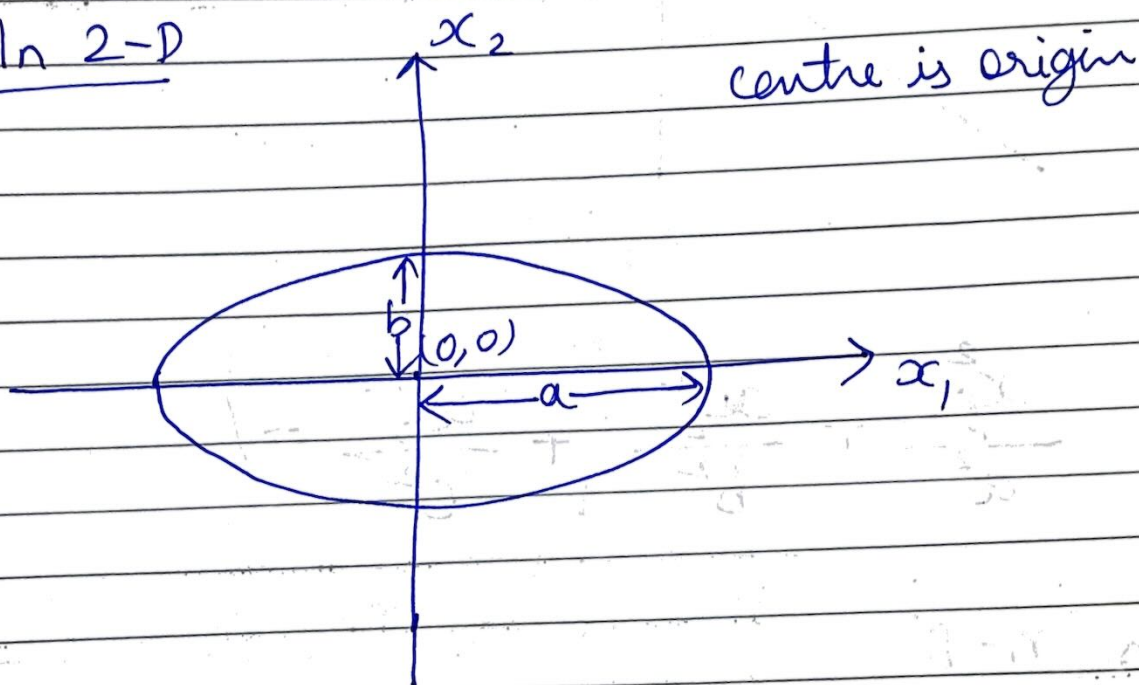
In n-D

$$x_1^2 + x_2^2 + \dots + x_n^2 = r^2$$

$$\Rightarrow \sum_{i=1}^n x_i^2 = r^2$$

V8: Equation of an Ellipse (2-D), Ellipsoid (3-D) and Hyperellipsoid (n-D)

In 2-D



$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\text{If } \frac{x_i^2}{a^2} + \frac{y_i^2}{b^2} < 1$$

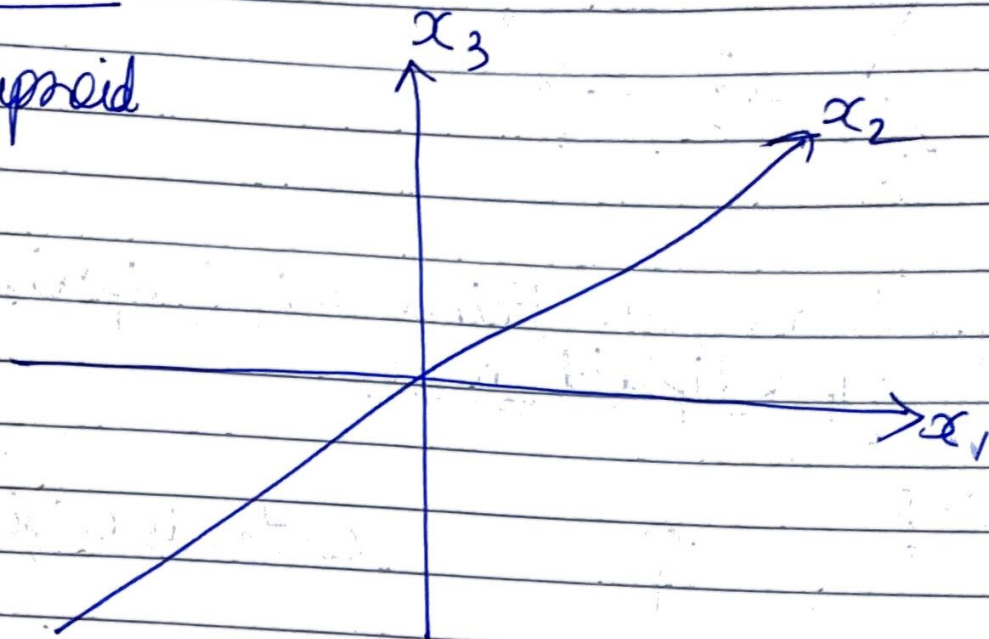
point lies
inside

> 1 point lies
outside

= on the ellipse

In 3-D

ellipsoid



$$\frac{x_1^2}{a^2} + \frac{x_2^2}{b^2} + \frac{x_3^2}{c^2} = 1$$

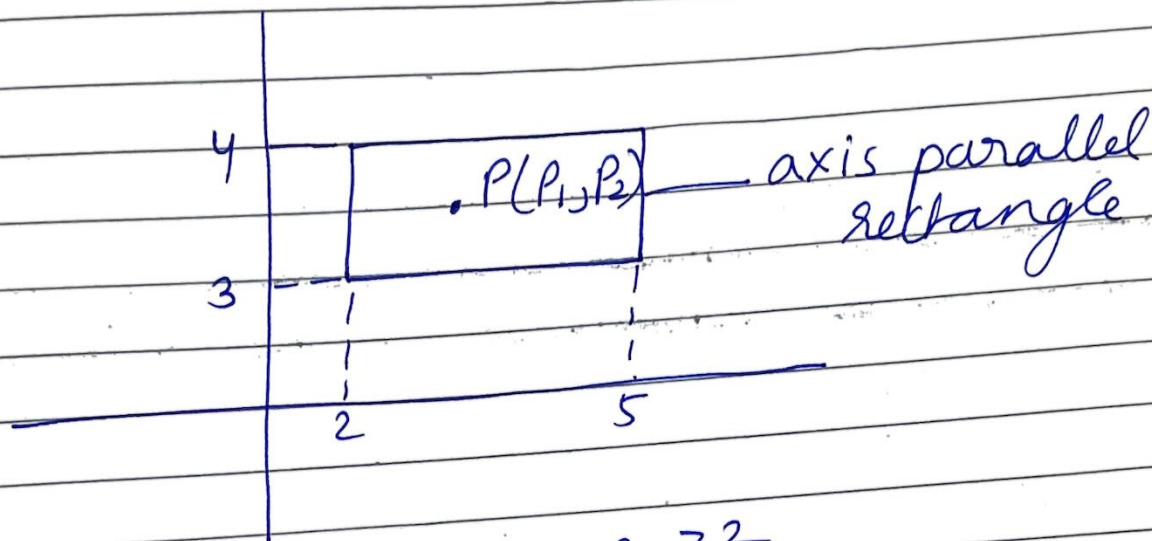
In n-D

$$\frac{x_1^2}{a^2} + \frac{x_2^2}{b^2} + \dots + \frac{x_n^2}{a_n^2} = 1$$

hyperellipsoid

$$\sum_{i=1}^n \frac{x_i^2}{a_i^2} = 1$$

V9: Square, Rectangle



if $P_1 < 5$ & $P_1 > 2$

if $P_2 > 3$ & $P_2 < 4$

then p lies inside

V10: Hypercube, HyperCuboid